CHAPTER 7

NATIONAL INCOME ACCOUNTING

CHAPTER ESSENCE AT A GLANCE:
Master circular flow of income, national income aggregates (GDP, GNP, NDP, NNP at factor cost and market price), three measurement methods, and GDP welfare limitations.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Circular Flow of Income:** In a two-sector economy without government or foreign trade, total output produced equals total factor income generated equals total consumption expenditure on goods.

• **Key Distinctions and Conversions:** Gross to Net requires deducting Depreciation; Domestic to National requires adding NFIA; Market Price to Factor Cost requires deducting Net Indirect Taxes (NIT = Indirect Taxes - Subsidies).

• **Three Measurement Methods:** Product or Value Added Method (GVA = Output - Intermediate Consumption); Income Method (Compensation of Employees + Operating Surplus + Mixed Income); Expenditure Method (C + I + G + Net Exports).

• **Real GDP vs Nominal GDP and Welfare:** Nominal GDP is valued at current prices while Real GDP is valued at constant base year prices; Real GDP is superior for growth comparisons, though GDP excludes non-monetary work, income distribution, and environmental externalities.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$\text{National Income Conversion} \iff \text{NNP}_{\text{FC}} = \text{GDP}_{\text{MP}} - \text{Dep} + \text{NFIA} - \text{NIT}$$

Fundamental Bridge from Gross Domestic to Net National

$$\text{Value Added Identity} \iff \text{GVA}_{\text{MP}} = \text{Value of Output} - \text{Intermediate Consumption}$$

Avoidance of Double Counting via Value Added

$$\text{GDP Expenditure Equation} \iff \text{GDP}_{\text{MP}} = C + I + G + (X - M)$$

Four Primary Expenditure Streams at Market Prices

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

Comprehensive Economy-Wide Price Index

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never include intermediate goods in National Income; doing so leads to double counting of intermediate inputs and overstates national output.

• **Alert:** Transfer payments (like pensions, student scholarships, gifts) must be strictly excluded from national income because they represent transfers without corresponding production.

• **Alert:** Depreciation is also called Consumption of Fixed Capital; Net Indirect Taxes equals Indirect Taxes minus Subsidies.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Final Goods and Intermediate Goods with one example of each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Final Goods:** Goods purchased for final consumption by households or capital investment by producers, crossing the production boundary with no further value added. Example: Bread bought by consumer, machine bought by firm. [1 Mark]

• **Intermediate Goods:** Goods used as raw material inputs for further production or resale within the same financial year. Example: Flour bought by a baker to make bread. [1 Mark]

2. Explain the circular flow of income in a simple two-sector economy with a neat diagrammatic description. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Two Sectors:** Households own factors of production (land, labour, capital, enterprise); Firms produce goods and services. [1 Mark]

• **Real Flow:** Households supply factor services to firms; firms produce and supply final goods and services to households. [1 Mark]

• **Money Flow:** Firms pay factor incomes (rent, wages, interest, profit) to households; households spend income buying goods from firms (Consumption Expenditure). Total Factor Income = Total Output = Total Expenditure. [1 Mark]

3. Explain the Expenditure Method of calculating Gross Domestic Product at Market Price (GDP_{MP}). List its components. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Concept:** Measures aggregate final expenditure incurred on goods and services produced within the domestic territory during a financial year. [1 Mark]

• **Components:**
• Private Final Consumption Expenditure (PFCE): Household spending on durable and non-durable goods and services.
• Government Final Consumption Expenditure (GFCE): State spending on public administration, education, defence. [1.5 Marks]

• • Gross Domestic Capital Formation (GDCF): Gross fixed capital formation + Change in stock (Closing stock - Opening stock).
• Net Exports (X - M): Export earnings minus import expenditure. Formula: GDP_{MP} = PFCE + GFCE + GDCF + (X - M). [1.5 Marks]

4. Calculate National Income (NNP_{FC}) from the following data (in Rs Crores): GDP_{MP} = 5,000; Consumption of Fixed Capital (Depreciation) = 400; Indirect Taxes = 600; Subsidies = 100; Net Factor Income from Abroad (NFIA) = -80. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Find Net Indirect Taxes):** NIT = Indirect Taxes - Subsidies = 600 - 100 = Rs 500 Crores. [1 Mark]

• **Step 2 (Deduct Depreciation):** NDP_{MP} = GDP_{MP} - Depreciation = 5,000 - 400 = Rs 4,600 Crores. [1 Mark]

• **Step 3 (Add NFIA):** NNP_{MP} = NDP_{MP} + NFIA = 4,600 + (-80) = Rs 4,520 Crores. [1 Mark]

• **Step 4 (Convert to Factor Cost):** NNP_{FC} = NNP_{MP} - NIT = 4,520 - 500 = Rs 4,020 Crores. National Income = Rs 4,020 Crores. [1 Mark]

5. Is GDP an adequate indicator of economic welfare? State any three limitations. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Limitation 1 (Distribution of GDP):** If GDP increases but concentrates in the hands of a few wealthy individuals, overall social welfare does not improve for the majority. [1 Mark]

• **Limitation 2 (Non-Monetary Exchanges):** Unpaid household work, childcare, and subsistence agriculture are excluded from GDP, understating actual production and welfare. [1 Mark]

• **Limitation 3 (Externalities):** Negative externalities like industrial pollution and deforestation reduce human health and living standards but are ignored in GDP computation. [1 Mark]

6. If Nominal GDP is Rs 12,000 Crores and Real GDP is Rs 10,000 Crores, calculate the GDP Deflator and interpret the result. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Calculation:** GDP Deflator = (Nominal GDP divided by Real GDP) * 100 = (12,000 divided by 10,000) * 100 = 120. **[1 Mark]**

• **Interpretation:** The general price level has increased by 20 percent compared to the base year price level. **[1 Mark]**